

Arnav Aggarwal

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SUMMARY

B.Tech. Computer Science student at Amity University Punjab, specialising in AI and full-stack development, with a track record of building and shipping real-world intelligent systems. Delivered an end-to-end AI-powered interview platform during a professional internship and achieved 98.4% accuracy on a published machine learning research project. Published researcher with work accepted at international conferences. Driven by a passion for solving meaningful problems at the intersection of AI, software engineering, and human experience.

EDUCATION

Amity University Punjab, Mohali

August 2023 – Present

B.Tech. in Computer Science & Engineering — CGPA: 7.59/10

G N Khalsa Secondary School, Sector 30B, Chandigarh

2023

Class XII (CBSE) — 69.2%

The Gurukul, Sector 20, Panchkula

2021

Class X (CBSE) — 90.2%

EXPERIENCE

Software and AI Intern — Amity University Punjab, Mohali

June – July 2025

- Designed and independently delivered ARIA (AI-Assisted Interview Assessment) — a full-stack, AI-powered interview automation platform — as the primary intern deliverable within a 2-month timeline.
- Worked across the full product lifecycle: system design, AI integration, real-time proctoring, session management, and automated scoring.

PROJECTS

ARIA — AI-Assisted Interview Assessment

- Built a full-stack React.js platform for AI-autonomous, resume-aware interviews aligned to job description and role seniority.
- Leveraged Google Gemini API and LangChain for dynamic question generation; integrated face-api.js for real-time gaze and presence monitoring.
- Managed stateful sessions with Redis and secured data via Firebase, enabling automated performance scoring and structured interview logging.

Real-Time Indian Sign Language Recognition System

- Built a real-time ISL recognition pipeline using MediaPipe, OpenCV, and XGBoost to classify hand gestures from webcam video streams.
- Designed a landmark-based feature extraction framework using 3D hand coordinates, normalization, and feature engineering for robust gesture recognition.
- Evaluated multiple ML/DL models including SVM, Random Forest, CNN, and MLP, achieving 98.4% accuracy with optimized inference and explainable AI-based feature analysis.

SKILLS

Languages: Python, C++, Java

Frameworks & Libraries: React.js, LangChain, MediaPipe, OpenCV, XGBoost

Tools & Platforms: Git, Firebase, Redis, Google Gemini API

Soft Skills: Problem-Solving, Communication, Time Management, Teamwork

PUBLICATION & ACHIEVEMENTS

- “Human-like Hiring at Scale: A Conversational AI and Computer Vision Approach” — **Published in IEEE Xplore Proceedings of MPCON 2026**, DOI: 10.1109/MPCON69668.2026.11508196
- “Explainable and Efficient ML Framework for Real-Time ISL Recognition Using Hand Landmark Features” — Submitted to IEEE Chandigarh Subsection International Conference 2026